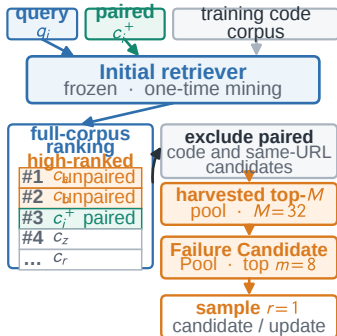


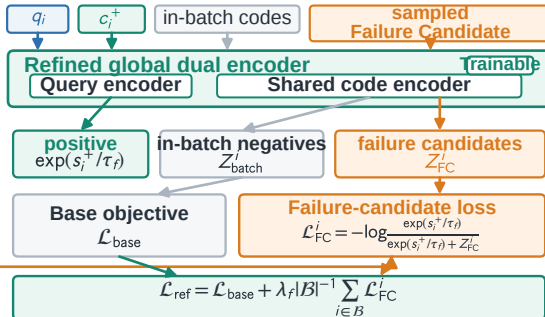
ReFCODE-Global · offline harvesting + global refinement

① Failure Candidate Harvesting



cached once · no dynamic re-mining

② Failure-Calibrated Contrastive Refinement



score-margin interpretation

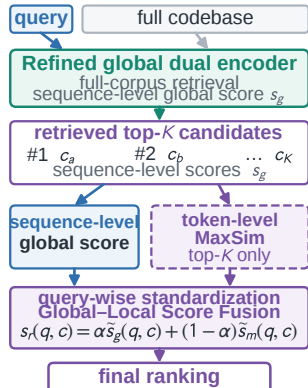
$$\mathcal{L}_{FC}^i = \log \left(1 + \sum_{c \in S_i} e^{-\mu_{\theta}(q_i, c)/\tau_f} \right)$$

small / negative margin
greater pressure

global representations only · no token-level training loss

ReFCODE-Rerank: Inference-time Extension

③ Dual-Granularity Reranking



MaxSim: no trainable parameters
inference latency + memory cost